



Unit:-II Supervised Learning-Regression (09 Hours)

Introduction to regression, need of regression, regression vs correlation.

Types of regression: Univariate vs Multivariate, Linear vs Nonlinear, Simple vs Multiple, Bias–Variance Tradeoff, Overfitting and Underfitting.

Regression Techniques: Simple and Multiple Linear Regression, Polynomial Regression, Decision Tree Regression, Random Forest Regression, Support Vector Regression.

Regularization Techniques: Ridge Regression (L2); Lasso Regression (L1).

Evaluation Metrics: Mean Squared Error (MSE); Mean Absolute Error (MAE); Root Mean Squared Error (RMSE); R-squared (R²).



Mapping of Course Objective and Outcome

Course Objective: The course aims to-

Develop an understanding of regression concepts, techniques, and evaluation metrics used for predictive modeling.

Course Outcome: Upon successful completion of this course, students will be able to-

CO2: Make use of various types of regression models for predictive modeling and data analysis.



Introduction to regression

Regression:

Regression is a **supervised machine learning technique** used to **predict a continuous numerical value** by learning the relationship between one or more independent variables (features) and a dependent variable (target).

Example

Using regression, if a new house has an area of **1300 sq.ft.**, the model can predict its approximate price (e.g., **₹65 Lakhs**).

House Area (sq.ft.)	House Price (₹ Lakhs)
800	40
1000	50
1200	60
1500	75



Need of regression

- **Predict continuous numerical values** such as price, temperature, or sales.
- **Analyze relationships** between dependent and independent variables.
- **Forecast future trends** using historical data.
- **Support data-driven decision-making** in business and research.
- **Estimate unknown values** for new or unseen data.
- **Optimize planning and resource allocation** by making accurate predictions.



Regression vs Correlation.

Regression	Correlation
Predicts the value of a dependent variable.	Measures the strength and direction of the relationship between variables.
Establishes a cause-and-effect (predictive) relationship.	Indicates only association , not causation.
Has dependent and independent variables.	Treats both variables equally , no dependent or independent variable.
Used for prediction and forecasting in Machine Learning.	Used for relationship analysis in statistics.

Example

- Correlation:** Examines whether **study hours** and **exam marks** are related.
- Regression:** Predicts **exam marks** based on the number of **study hours**.



Types of regression: Univariate vs Multivariate.

Univariate Regression	Multivariate Regression
Uses one independent variable (feature) to predict the target variable.	Uses two or more independent variables (features) to predict the target variable.
Simpler model with lower computational complexity.	More complex model that captures multiple influencing factors.
Suitable when the output depends on a single feature.	Suitable when the output depends on multiple features.
Example: Predict house price using house area only.	Example: Predict house price using area, number of bedrooms, location, and age of the house.

Key Point

- **Univariate Regression:** One input feature → One output.
- **Multivariate Regression:** Multiple input features → One output.



Types of regression: Linear

- Linear regression is one of the most fundamental machine learning algorithms used for predicting continuous values. It establishes a **relationship** between **independent variables** ($\bar{x}$) and a **dependent variable** ($\bar{y}$) using a **straight-line equation**.

Equation of Linear Regression

The equation of simple linear regression (with one independent variable) is:

$$Y = mX + c$$

where:

Y = Dependent variable (target)

X = Independent variable (input feature)

m = Slope of the line (coefficient)

c = Intercept (constant)



Types of regression: Linear

Step 1: Collect Data

Suppose we want to predict a student's exam score based on their study hours.

Step 2: Visualize the Data

Before applying regression, plot the data points to check if there is a linear relationship. If the data follows an increasing or decreasing trend, linear regression is a good choice.

Step 3: Calculate the Regression Line

The regression line equation is: $Y = mX + c$

We calculate **m(slope)** and **c (intercept)** using the formulas:

$$m = (n\sum XY - (\sum X)(\sum Y)) / (n\sum X^2 - (\sum X)^2)$$

Using our data: $c = (\sum Y - m\sum X) / n$

- $\sum X = 1 + 2 + 3 + 4 + 5 = 15$
- $\sum Y = 50 + 55 + 65 + 70 + 75 = 315$
- $\sum XY = (1 \times 50) + (2 \times 55) + (3 \times 65) + (4 \times 70) + (5 \times 75) = 1105$
- $\sum X^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$
- $n = 5$

Study Hours (X)	Exam Score (Y)
1	50
2	55
3	65
4	70
5	75



Types of regression: Linear

Now calculating m:

$$m = ((5 \times 1105) - (15 \times 315)) / ((5 \times 55) - (15^2))$$

$$m = (55525 - 4725) / (275 - 225) = 800 / 50 = 16$$

Now calculating c:

$$c = (315 - (16 \times 15)) / 5$$

$$c = (315 - 240) / 5 = 75 / 5 = 15$$

So, the final equation is: **$Y = 16X + 15$**

Step 4: Make Predictions

Now, let's predict the exam score if a student studies for **6 hours**:

$$Y = 16(6) + 15 = 96$$

So, a student who studies for **6 hours** is predicted to score **96 marks**.



Types of regression: Nonlinear(Polynomial)

Nonlinear Regression is a supervised machine learning technique used to model the relationship between input and output variables when the relationship is **curved rather than a straight line**.

General Equation

The most common nonlinear regression model is **Polynomial Regression**.

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \dots + \beta_n X^n$$

Where:

- **Y** = Dependent (target) variable
- **X** = Independent (input) variable
- **β_0** = Intercept
- **$\beta_1, \beta_2, \dots, \beta_n$** = Regression coefficients
- **n** = Degree of the polynomial

Example : Given the nonlinear regression equation:

$$Y = 2 + 3X + X^2$$

Predict the value of **Y** when **X = 4**.



Types of regression: Nonlinear(Polynomial)

Step 1: Write the Equation

$$Y = 2 + 3X + X^2$$

Step 2: Substitute the Value of X

Let

$$X = 4$$

Substitute into the equation:

$$Y = 2 + 3(4) + (4)^2$$

Step 3: Perform the Calculations

$$Y = 2 + 12 + 16$$

Step 4: Find the Result

$$Y = 30$$

Real-Life Example

Crop Yield Prediction

- Input (X):** Amount of fertilizer (kg/hectare)
- Output (Y):** Crop yield (tons/hectare)

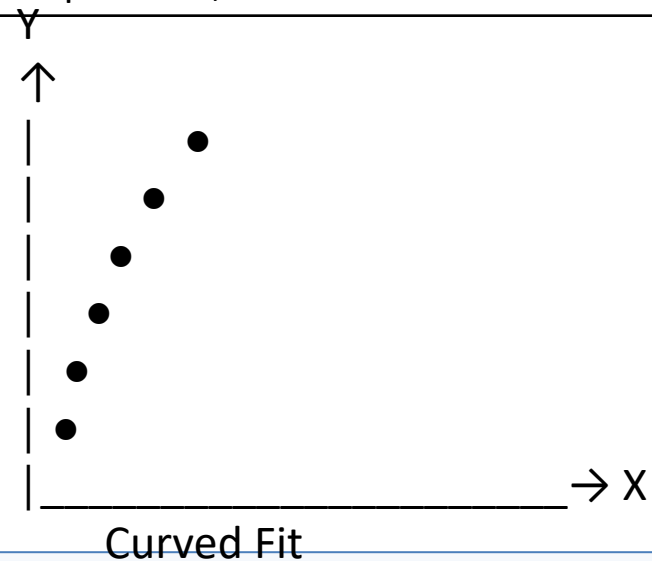
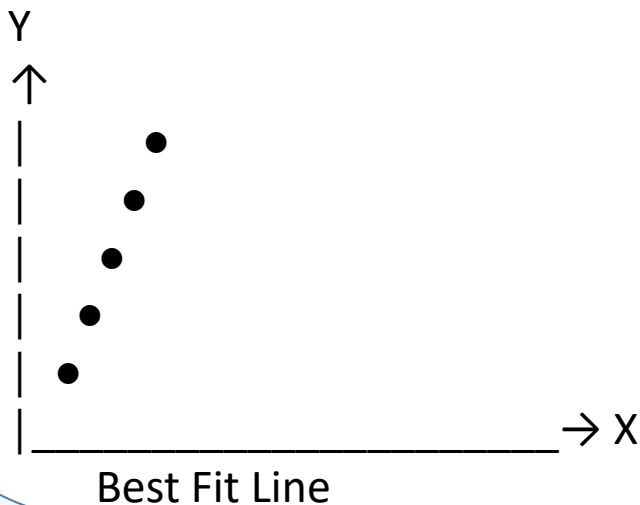
Initially, increasing fertilizer increases crop yield. However, after an optimal level, additional fertilizer has little effect or may even reduce yield.

This curved relationship is better modeled using **nonlinear regression** than a straight-line model.



Types of regression: Linear vs Nonlinear

Linear Regression	Nonlinear Regression
Assumes a linear (straight-line) relationship between input and output variables.	Assumes a nonlinear (curved) relationship between input and output variables.
Simple to understand, implement, and interpret.	More complex and can model intricate relationships.
Best suited for data with a linear trend .	Best suited for data with curved or complex patterns .
Example: Predict salary based on years of experience.	Example: Predict crop yield based on rainfall, temperature, and soil conditions.





Types of regression: Simple vs Multiple

Simple Linear Regression	Multiple Linear Regression
Uses one independent variable (X) to predict one dependent variable (Y).	Uses two or more independent variables ($X_1, X_2, \dots, X_n$) to predict one dependent variable (Y).
Models the relationship with a single predictor.	Models the relationship using multiple predictors simultaneously.
Easy to understand and implement.	More accurate for complex real-world problems but computationally more complex.
Example: Predict salary using years of experience.	Example: Predict house price using area, bedrooms, location, and age of the house.



Bias–Variance Tradeoff, Overfitting and Underfitting.

Bias–Variance Tradeoff is the process of finding the optimal balance between **bias (error due to overly simple models)** and **variance (error due to overly complex models)** to achieve the best prediction performance on unseen data.

- **Bias** is the error caused by making overly simple assumptions in the learning algorithm.
- High bias leads to **underfitting**, where the model cannot capture the underlying pattern of the data.

Characteristics:

Simple model ,Low training accuracy ,High prediction error ,Poor generalization

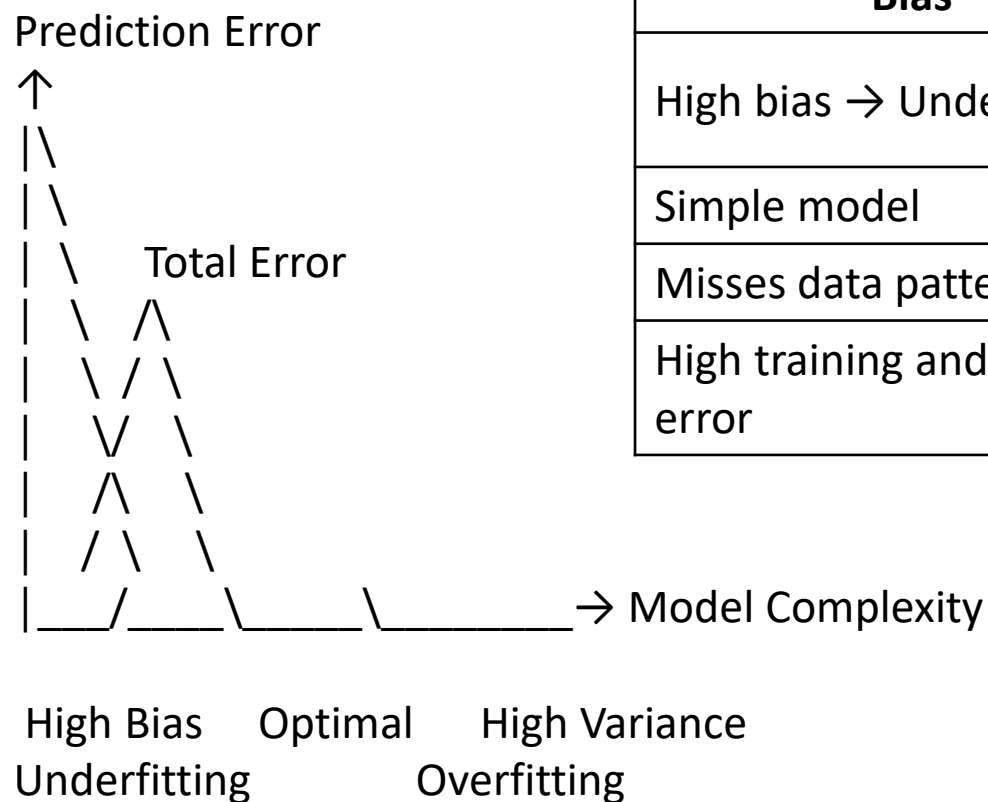
- **Variance** is the error caused by excessive sensitivity to the training data.
- High variance leads to **overfitting**, where the model learns the training data too closely, including noise.

Characteristics:

Complex model ,High training accuracy ,Poor performance on new data ,Sensitive to small data changes



Bias–Variance Tradeoff, Overfitting and Underfitting



Bias	Variance
High bias → Underfitting	High variance → Overfitting
Simple model	Complex model
Misses data patterns	Learns noise in data
High training and testing error	Low training error but high testing error



Bias–Variance Tradeoff, Overfitting and Underfitting

Example

Suppose we want to predict **house prices**.

High Bias (Underfitting):

- Model: $Y = 50 + 2X$
- Uses **only house area**.
- **Ignores other factors** like location and number of bedrooms.
- Predictions are inaccurate.

High Variance (Overfitting):

- Model **uses area, bedrooms, age, location, parking, floor, paint color, etc.**
- Fits the training data almost perfectly.
- **Performs poorly on new house data.**

Balanced Model:

- Uses the **most relevant features** (e.g., area, location, bedrooms).
- Produces accurate predictions on both training and unseen data.

High Bias → Underfitting

High Variance → Overfitting

Balanced Bias and Variance → Best Generalization

The goal is to minimize the **total prediction error**.



Decision Tree Regression

Decision Tree Regression is a **supervised machine learning** algorithm that predicts **continuous numerical values** by recursively splitting the dataset into smaller subsets based on feature values. The final prediction is the **average value of the target variable** in the leaf node.

Working Principle:

1. Start with the entire dataset.
2. Select the best feature and split point that minimizes prediction error (usually Mean Squared Error).
3. Divide the dataset into smaller subsets.
4. Repeat the process recursively until a stopping condition is met.
5. Predict the output using the **average target value** of the leaf node.

Example:

Step 1: Build the Decision Tree

Split based on **House Area = 1200 sq.ft.**

- **Area ≤ 1200** $\rightarrow \{40, 50, 60\}$
- **Area > 1200** $\rightarrow \{75, 90\}$

House Area (sq.ft.)	Price (₹ Lakhs)
800	40
1000	50
1200	60
1500	75
1800	90



Decision Tree Regression

Step 2: Calculate Leaf Node Values

For $\text{Area} \leq 1200$

$$\frac{40 + 50 + 60}{3} = 50$$

Prediction = ₹50 Lakhs

For $\text{Area} > 1200$

$$\frac{75 + 90}{2} = 82.5$$

Prediction = ₹82.5 Lakhs

Step 3: Prediction

Suppose,

House Area = 1700 sq.ft.

Since

1700 > 1200

The prediction belongs to the second leaf.

Predicted House Price = ₹82.5 Lakhs

Advantages:

Handles **linear and nonlinear** relationships.

No need for feature scaling or normalization.

Easy to understand and visualize.

Works with both numerical and categorical features.

Captures complex decision boundaries.

Disadvantages:

Can **overfit** the training data if the tree is too deep.

Sensitive to small changes in the dataset.

Less accurate than ensemble methods like Random Forest for many problems.



Random Forest Regression

Random Forest Regression is a **supervised machine learning algorithm** that predicts **continuous numerical values** by combining the predictions of **multiple Decision Trees**. The final prediction is obtained by taking the **average of all tree predictions**, resulting in higher accuracy and reduced overfitting.

Working Principle

1. Draw multiple **bootstrap samples** (random samples with replacement) from the training dataset.
2. Train a **Decision Tree** on each sample.
3. At each split, consider a **random subset of features**.
4. Each tree predicts a numerical value.
5. The **final prediction** is the **average** of all tree predictions.

Example:



Random Forest Regression

Advantages:

- Higher prediction accuracy than a single Decision Tree.
- Reduces **overfitting** by averaging multiple trees.
- Handles **linear and nonlinear** relationships effectively.
- Works well with large datasets and high-dimensional data.
- Robust to noise and missing values.

Disadvantages:

- More computationally expensive than a single Decision Tree.
- Requires more memory and training time.
- Less interpretable due to the use of many trees.

Applications

- House price prediction
- Crop yield prediction
- Stock price forecasting
- Sales forecasting
- Used car price estimation
- Medical cost prediction



Support Vector Regression.

Support Vector Regression (SVR) is a **supervised machine learning algorithm** used to predict **continuous numerical values**. Unlike Linear Regression, SVR tries to find the **best-fit line (or hyperplane)** that keeps most data points within a specified error margin called ϵ (**epsilon**) while maximizing the margin.

Working Principle

1. Collect the training dataset.
2. Select the **epsilon (ϵ)** value (acceptable error).
3. Find the regression line (or hyperplane) that fits the data within the ϵ -margin.
4. Identify the **Support Vectors** (points outside or on the margin).
5. Use the regression model to predict the output for new data.

Key Idea of SVR

- SVR predicts **continuous values**.
- It does **not** try to minimize the error for every data point.
- Instead, it allows a small error (ϵ) and focuses on fitting the data within this margin.
- Data points outside the ϵ -margin are called **Support Vectors**, and they determine the regression model.

Example:



Support Vector Regression.

Advantages:

- Effective for **linear and nonlinear** regression problems.
- Produces good predictions even with small datasets.
- Handles high-dimensional data efficiently.
- Robust to outliers due to the ϵ -insensitive loss function.

Disadvantages:

- Computationally expensive for very large datasets.
- Choosing the right **kernel** and parameters (**C**, **γ** , **ϵ**) can be challenging.
- Slower training compared to Linear Regression.

Applications

- House price prediction
- Stock market forecasting
- Crop yield estimation
- Weather forecasting
- Sales prediction
- Medical cost prediction



Regularization Techniques: Ridge Regression (L2), Lasso Regression (L1).

Regularization is a technique used in machine learning to **reduce overfitting** by adding a **penalty term** to the loss (cost) function. It discourages the model from assigning excessively large coefficients to features, thereby improving the model's ability to generalize to new data.

Need of Regularization:

- Reduces overfitting
- Improves model generalization
- Prevents large coefficient values
- Handles multicollinearity
- Produces more reliable predictions

Ridge Regression (L2 Regularization):

Ridge Regression is a regularization technique that adds the **sum of the squares of the coefficients (L2 penalty)** to the cost function. It **shrinks** the coefficients toward zero but **does not make them exactly zero**.

Example:

Cost Function

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

Where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- β_j = Regression coefficient
- λ = Regularization parameter



Regularization Techniques: Ridge Regression (L2), Lasso Regression (L1).

Lasso Regression (L1 Regularization):

Lasso Regression is a regularization technique that adds the **sum of the absolute values of the coefficients (L1 penalty)** to the cost function. It can shrink some coefficients **exactly to zero**, thereby performing **feature selection**.

Cost Function

$$J(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

$y_i \rightarrow$ Actual value

$\hat{y}_i \rightarrow$ Predicted value

$n \rightarrow$ Number of training samples

$p \rightarrow$ Number of input features

$\beta_j \rightarrow$ Feature coefficient (weight)

$\lambda \rightarrow$ Controls the amount of regularization

L1 Penalty ($\sum |\beta_j|$) $\rightarrow$ Shrinks coefficients and can set some coefficients **exactly to zero**, performing **automatic feature selection**.

Example:



Evaluation Metrics

- **Mean Squared Error (MSE):**

Mean Squared Error (MSE) is the **average of the squared differences** between the actual and predicted values.

Formula

$$MSE = \frac{\sum (Y - \hat{Y})^2}{n}$$

Where

- **Y** = Actual value **Formula**

- **$\hat{Y}$** = Predicted value

- **n** = Number of samples

$$MAE = \frac{\sum |Y - \hat{Y}|}{n}$$

- **Mean Absolute Error (MAE):**

Mean Absolute Error (MAE) is the **average of the absolute differences** between actual and predicted values

Formula

$$MAE = \frac{\sum |Y - \hat{Y}|}{n}$$



Evaluation Metrics

- **Root Mean Squared Error (RMSE):**

Root Mean Squared Error (RMSE) is the **square root of the Mean Squared Error**. It measures the prediction error in the **same unit as the target variable**.

Formula

$$RMSE = \sqrt{MSE}$$

- **R-squared (R²):**

R² (Coefficient of Determination) measures the proportion of the variance in the dependent variable that is explained by the regression model.

Formula

$$R^2 = 1 - \frac{\sum (Y - \hat{Y})^2}{\sum (Y - \bar{Y})^2}$$

Where

- **Y** = Actual value
- **$\hat{Y}$** = Predicted value
- **$\bar{Y}$** = Mean of actual values



Case Study

Comparative Study of Regression Techniques for House Price Prediction